

Shattered Bytes

A Manual Data Carving Serious Game for Computer Forensics Education

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What the game trains

Shattered Bytes focuses on the complete **manual data carving workflow**: detecting meaningful byte patterns, rejecting false leads, reconstructing fragmented evidence, and reporting only what can be defended from the recovered data.

- The player must combine **signature analysis**, **offset reasoning**, and **fragment reconstruction**.
- The campaign avoids a pure “find the header” exercise by adding decoys, partial recovery, XOR obfuscation, and partition-based navigation.
- Recent gameplay additions include a drag-and-drop **evidence intake** sequence and a final **incident timeline board**.
- The final output is both the recovered evidence and a transparent **competency report** on the player’s investigative method.

Teaching problem

Data carving can look like simple **signature matching**, but real recovery also requires boundary validation, fragment ordering, decoy rejection, and careful reporting.

Design answer

The player works directly on **raw bytes**, selects exact ranges, reconstructs evidence, verifies the result, and submits a forensic conclusion.

The game turns a low-level forensic task into a sequence of **explicit investigative decisions**.

Operation Shattered Meridian

A regional hospital network has been hit by the Night Meridian ransomware cell. The live breach is contained, but the payment channel, mule identities, ATM cash-out, fraudulent transfer, ransomware payload, and exfiltration path are still hidden in seized devices.

- **Player role:** junior forensic analyst assigned to Agent Root.
- **Mission:** reconstruct six digital evidence items from raw dumps.
- **Final result:** one defensible incident chain.

Core Gameplay Loop

- ① **Prepare** the evidence workflow: sealed drive, write blocker, forensic workstation, hash verification.
- ② **Identify** the forensic target from the mission briefing.
- ③ **Inspect** the dump with search, go, select, and info.
- ④ **Stash** exact byte ranges in the workbench.
- ⑤ **Compose** fragments, apply XOR when needed, and carve the artifact.
- ⑥ **Verify** the recovered image or text payload in the asset viewer.
- ⑦ **Report**: recovered, partial, or inconclusive.

Design principle

The player is rewarded for a disciplined forensic workflow: observe, validate, reconstruct, and **report only what the bytes support**.

Campaign Progression

Act	Topic	Main forensic skill	Outcome
1	False PNG lead	Header/footer validation	Full recovery
2	Fragmented PNG	Descriptor-based reconstruction	Full recovery
3	Multi-signature triage	Relevance over recognisable files	Full recovery
4	MBR partition	Little-Endian LBA to byte offset	Full recovery
5	Obscured tail	XOR known-plaintext and partial recovery	Partial report
6	Ransomware aftermath	Multi-fragment XOR recovery	Full recovery

- **Hex Editor:** byte-level inspection and range selection.
- **Terminal:** forensic commands such as `search`, `go`, `select`, `entropy`, `xorcalc`, `xor`, and `report`.
- **Workbench:** fragment storage, ordering, XOR transformation, carving.
- **Asset Viewer:** visual/text verification of recovered evidence.
- **Evidence Intake:** drag-and-drop acquisition procedure before Act 1.
- **Timeline Board:** final drag-and-drop correlation of all recovered artefacts.
- **Objectives Panel:** explicit investigation goals and manual hints.

Score formula

```
score = maxScore - 15*hints - 25*badSelections - 30*extraCarves -  
40*wrongReports - 15*procedureMistakes + timeBonus
```

- **Bad selections:** decoys, unsupported ranges, overlaps, or extra bytes.
- **Extra carves:** carving before validating boundaries and order.
- **Wrong reports:** overclaiming partial evidence or misclassifying recovery.
- **Procedure mistakes:** failed post-action forensic decisions.
- **Final mastery:** signature recognition, offset precision, fragment reconstruction, trial-and-error discipline, obfuscation handling, procedure discipline, forensic reasoning, reporting quality.

Technical Implementation

- **Frontend:** React and Vite.
- **Levels:** data-driven JSON dumps in `public/levels/`.
- **Generators:** deterministic Python scripts in `scripts/level_generation/`.
- **Evidence:** source artefacts and UI visuals in `assets/`; playable level data in `public/levels/`.
- **Media:** intro video in `public/intro.mp4`, with browser-safe audio enable.
- **Runtime:** browser-based interaction, no backend dependency during play.

Reproducibility

The campaign levels can be regenerated with `npm run levels:build`; the project can be built with `npm run build`. This keeps the game **data-driven** and reproducible.

Course alignment

- **file signatures** and data carving;
- fragmentation and unallocated space;
- MBR and **Little-Endian** values;
- anti-forensics and decoys;
- partial recovery and defensible reporting.

Suggested demo

Show the **evidence intake**, Act 1 for validation, Act 4 for offset reasoning, Act 5 or 6 for XOR, then close with the **timeline board** and final competency report.

Questions?